

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 22****Topics covered :**

1. Effect of time-shift and Derivative on Fourier Coefficients
2. Convergence of Fourier Series

- The Complex Fourier Coefficients of an even function are purely real.

Proof: Since $f(x)$ is real, $C_{-k} = C_k^*$. Further, since $f(x)$ is even,

$$C_{-k} = \frac{1}{L} \int_0^L f(x) e^{-j(-k)\omega_0 x} dx = \frac{1}{L} \int_0^L f(x) e^{jk\omega_0 x} dx \quad (1)$$

Using $x = -u$, (1) can be modified as

$$C_{-k} = \frac{1}{L} \int_0^{-L} \underbrace{f(-u)}_{=f(u)} e^{-jk\omega_0 u} (-du) \quad (2)$$

Further, using $v = u + L$, (2) can be modified as

$$C_{-k} = \frac{1}{L} \int_0^L \underbrace{f(u+L)}_{=f(u)} e^{-jk\omega_0 v} \underbrace{e^{-jk\omega_0 v}}_1 (du) = C_k \quad (3)$$

Thus for real and even signals $C_{-k} = C_k^*$ and $C_{-k} = C_k$, which requires C_k to be real. In a similar way, it can be shown that the Fourier Coefficients of an odd function are purely imaginary.

- **Impact of Time Shift/Delay on Fourier Coefficients:** Let C_k denote the Fourier coefficients of a signal $x(t)$. The Fourier coefficients, denoted by C'_k , of a time-delayed signal $x(t - \tau)$ can be derived as follows

$$\begin{aligned} C'_k &= \frac{1}{T} \int_0^T x(t - \tau) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\tau}^{T-\tau} x(t') e^{-jk\omega_0 (t'+\tau)} dt \\ &= e^{-jk\omega_0 \tau} \frac{1}{T} \int_{-\tau}^{T-\tau} x(t') e^{-jk\omega_0 (t'+\tau)} dt = e^{-jk\omega_0 \tau} C_k \end{aligned} \quad (4)$$

This result suggests that the time shift does not alter the magnitude of the Fourier coefficients but introduces a phase delay instead.

- **Effect of Derivative on Fourier Coefficients:** If C_k represents the Fourier coefficients of a signal $x(t)$ i.e.,

$$x(t) = \sum_{-\infty}^{\infty} C_k e^{jk\omega_0 t} \quad (5)$$

Then the Fourier coefficients (say C'_k) of a signal that is the derivative of the original signal $x(t)$ can be computed

using synthesis equation as

$$y(t) = \dot{x}(t) = \sum_{-\infty}^{\infty} (jk\omega_0 C_k) e^{jk\omega_0 t} \quad (6)$$

The Fourier coefficients of the signal $y(t)$ is thus

$$C'_k = jk\omega_0 C_k \quad (7)$$

- **Pointwise Convergence:** Let $s_N(t)$ represent the signal synthesized using the Fourier coefficients from $-N$ to N of the original signal $x(t)$. Then, $s_N(t)$ converges to $x(t)$ at all points, except at points of discontinuity, if the following conditions, commonly known as the Dirichlet conditions, are satisfied:

1. $x(t)$ is bounded.
2. $x(t)$ has a finite maximum and minimum within one period.
3. $x(t)$ has a finite number of discontinuities within one period.

At points of discontinuity, the synthesized signal approaches the average of the left-hand and right-hand limits of the function at that discontinuity.